Report

```python
import os
from pathlib import Path
import duckdb
import pandas as pd

_project_root = Path.cwd()
while not (_project_root / ".git").exists():
    _project_root = _project_root.parent
    if _project_root == _project_root.parent:
        raise RuntimeError(".git not found - are you inside the repo?")
os.chdir(_project_root)
print(os.getcwd())
```

```
/Users/isaacwedaman/Documents/CS_Stuff/iwedaman-okemo-data
```

Here, we are loading in all the parquets we have saved to do some data exploration on.

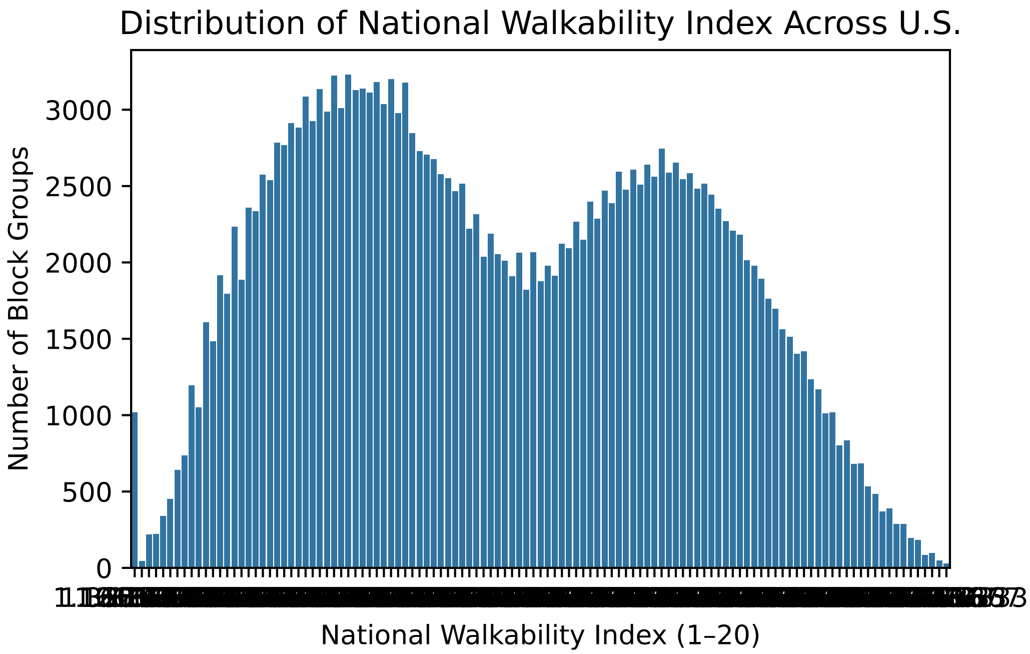
```python
import duckdb
con = duckdb.connect()
for parquet in ["metro", "walk_metrics", "geography"]:
    con.execute(f"""
      CREATE OR REPLACE VIEW {parquet} AS
      SELECT * FROM read_parquet('Data/processed/{parquet}.parquet')
    """)
```

Exploring the data. here, we see two peaks, with 9-13 and 13 - 16 being the most frequent walkability scores out of 20

```python
#merging both
# SQL
import matplotlib.pyplot as plt
import seaborn as sns




#reusing my old code jsut to have NatWalkIndexDist around
#here is generating walkability
con.execute(f"""
    CREATE OR REPLACE VIEW walkability AS
    SELECT * FROM read_parquet('{"Data/raw/walkability.parquet"}')
""")
print(con.execute("SELECT COUNT(*) AS n_blockgroups FROM walkability").df())
#here is getting cleaned
con.execute("""
    CREATE OR REPLACE VIEW cleaned AS
    SELECT
        *,
        LPAD(CAST(GEOID20 AS BIGINT)::VARCHAR, 12, '0')               AS geoid20_str,
        SUBSTR(LPAD(CAST(GEOID20 AS BIGINT)::VARCHAR, 12, '0'),1,5)   AS county_fips,
        LPAD(CAST(STATEFP AS BIGINT)::VARCHAR, 2, '0')               AS state_fips,
        NULLIF(D4A, -99999) AS D4A_clean
    FROM walkability
""")
con.execute("SELECT geoid20_str, county_fips, state_fips FROM cleaned LIMIT 5").df()



NatWalkIndexDist = con.execute("""
    SELECT NatWalkInd, COUNT (*) AS row_count FROM cleaned
        GROUP BY NatWalkInd
        ORDER BY NatWalkInd ASC
""").df()



sns.barplot(data=NatWalkIndexDist, x="NatWalkInd", y = "row_count")
plt.title("Distribution of National Walkability Index Across U.S.")
plt.xlabel("National Walkability Index (1-20)")
plt.ylabel("Number of Block Groups")
```

```
   n_blockgroups
0         220740
```

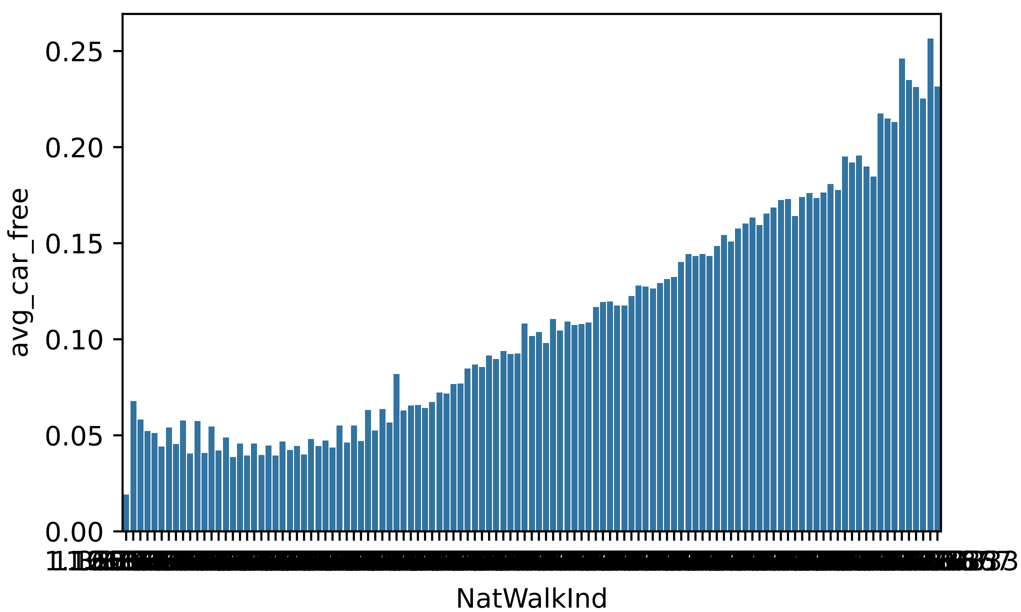
```
Text(0, 0.5, 'Number of Block Groups')
```



Something of note here is that we have made a view already - we can reference walk_metrics, no need to add the .parquet

```
walkindex_percent = con.execute("""
    SELECT NatWalkInd, AVG(PCT_AO0) as avg_car_free, COUNT (*) AS n FROM walk_metrics
        GROUP BY NatWalkInd
        ORDER BY NatWalkInd ASC
""").df()
print(walkindex_percent.describe())

#now the pandas evil twin
# print(walkindex_percent.groupby("NatWalkInd")["avg_car_free"].mean())
wm = pd.read_parquet("Data/processed/walk_metrics.parquet")
print(wm.groupby("NatWalkInd")["Pct_AO0"].mean())

sns.barplot(data=walkindex_percent, x="NatWalkInd", y = "avg_car_free")
```

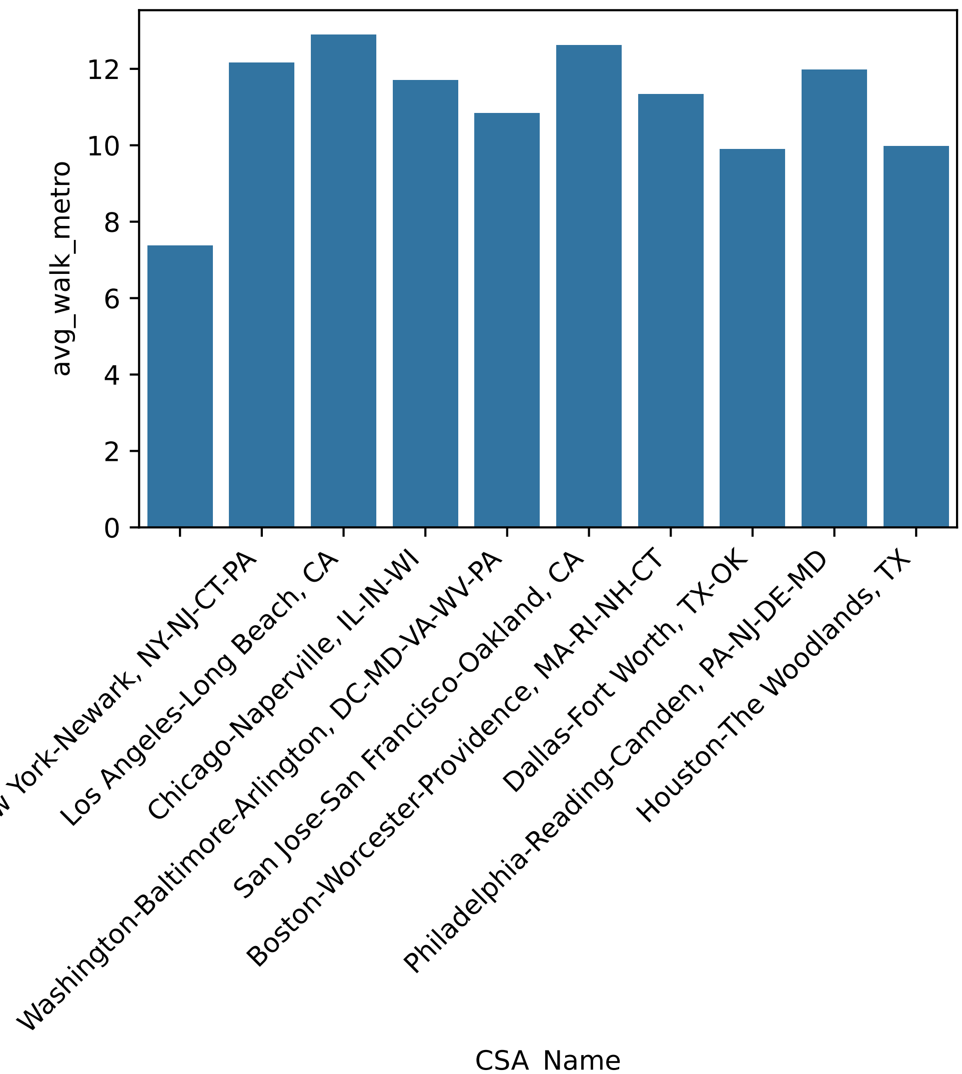
```
       NatWalkInd  avg_car_free            n
count  115.000000    115.000000   115.000000
mean    10.500000      0.108949  1919.478261
std      5.556944      0.059278   937.550432
min      1.000000      0.019025    29.000000
25%      5.750000      0.054766  1212.500000
50%     10.500000      0.101638  2179.000000
75%     15.250000      0.155935  2584.500000
max     20.000000      0.256497  3229.000000
NatWalkInd
1.000000     0.019025
1.166667     0.067674
1.333333     0.058184
1.500000     0.052134
1.666667     0.051081
               ...
19.333333    0.234834
19.500000    0.231154
19.666667    0.225232
19.833333    0.256497
20.000000    0.231420
Name: Pct_AO0, Length: 115, dtype: float64
```



```
#ill worry about this later - sanity check to remember it
print(pd.read_parquet("Data/processed/metro.parquet").columns.tolist())
```

```
['TotPop', 'geoid20_str', 'CSA', 'CSA_Name', 'NatWalkInd']
```

```
new_sql = con.execute("""
  SELECT CSA_Name, SUM(TotPop) AS metro_pop, AVG(NatWalkInd) AS avg_walk
  FROM metro
  WHERE CSA_Name IS NOT NULL
  GROUP BY CSA_Name
  ORDER BY metro_pop DESC
  LIMIT 10
""").df()


#pandas evil twin
metro_df = pd.read_parquet("Data/processed/metro.parquet")
metro_df.groupby("CSA_Name").agg(metro_pop=("TotPop","sum"), avg_walk_metro=("NatWalkInd","me

top10_metros = metro_df.groupby("CSA_Name").agg(metro_pop=("TotPop","sum"), avg_walk_metro=("

sns.barplot(data=top10_metros, x="CSA_Name", y = "avg_walk_metro")
plt.xticks(rotation=45, ha='right')
plt.show()
```



What did I learn? Well, especially if I am factoring my most recent discovery to be the most informative, that i have to do way more research. No way that New york is half as walkable as Los Angeles, and considerably less walkable than houston (I watched a youtube video about this). I do really like my previous “walk score vs ar ownership” correlation graph, however. my original goal for this project was to correlate more walkable cities, counties, and states with economic trends and municipality expenditures, with the goal of informing zoning and even cultural change! Ill keep digging!